



Smart Insurance Policy Recommendation and Claim Management System

CSA0905 – Programming In Java

**Guided by:
Dr. Sivasankar C**

Presented By:
R. Indhu (192524332)
A. Lakshmi Gayathri (192411188)
S. Karanya(192421309)



Engineer to Excel

Review Feedback



Feedback Received	Corrective Actions
Include real-time insurance and customer data	Integrated real-time inputs such as weather conditions, vehicle usage, claim status, and policy updates through APIs
Use suitable datasets for AI model development	Added insurance claim, customer profile, vehicle, policy, and fraud-related datasets for model training and testing
Make policy recommendation more intelligent	Developed an AI-based personalized recommendation engine that suggests policies based on customer risk profile, coverage needs, budget, and past claims
Improve claim management using AI	Added AI-based claim risk assessment, document analysis, and automated claim prioritization to identify genuine and suspicious claims
Introduce innovative fraud detection	Implemented an AI anomaly-detection model to identify unusual claim patterns, duplicate claims, abnormal claim amounts, and suspicious customer behavior
Improve real-time decision making	Added a dynamic risk score that continuously updates using real-time data and AI predictions, enabling faster policy and claim decisions

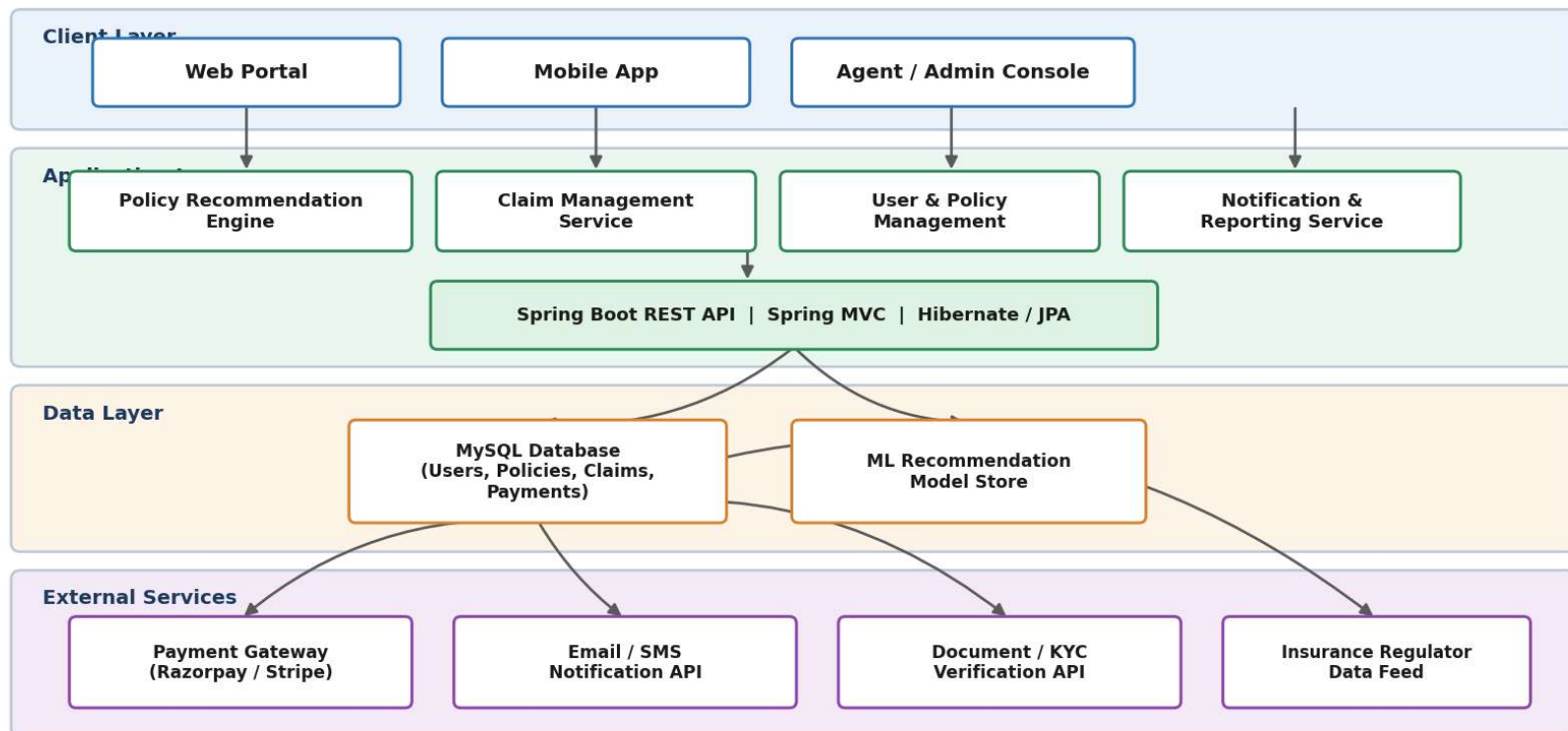


Engineer to Excel



Updated Architecture

Smart Insurance Policy Recommendation & Claim Management — Architecture



Module Design

Module	Purpose	Data Flow	Interface
Customer Management	Manage customer details and risk data	Customer → Database → AI	Registration, Profile
Policy Recommendation	Recommend suitable policies using AI	Customer + Dataset + Real-time Data → AI	Suggestions, Comparison
Claim Processing	Process claims and detect fraud using AI	Claim → AI Analysis → Database	Submission, Tracking, Status

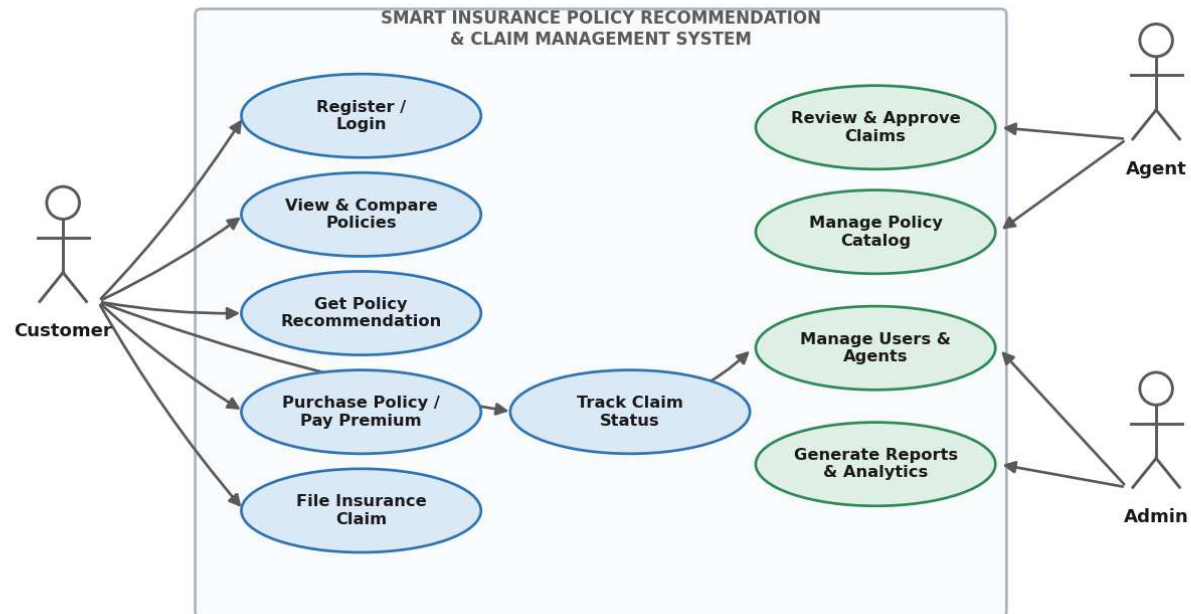


Engineer to Excel

Use Case Diagram



Smart Insurance System — Use Case Diagram



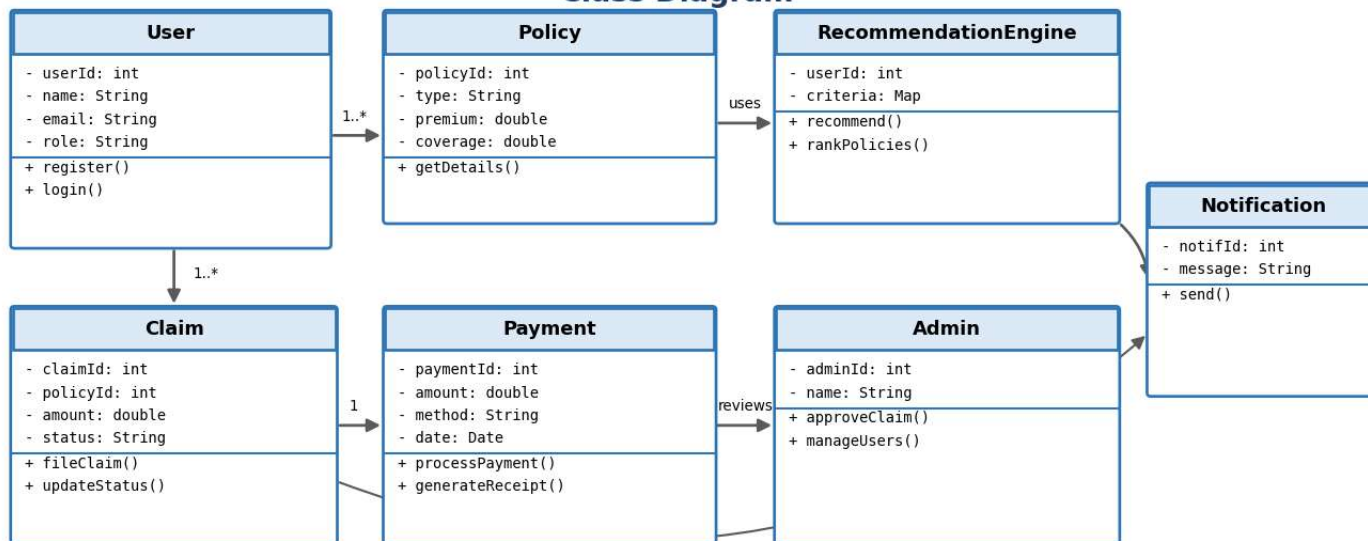


Engineer to Excel



Class Diagram

Class Diagram



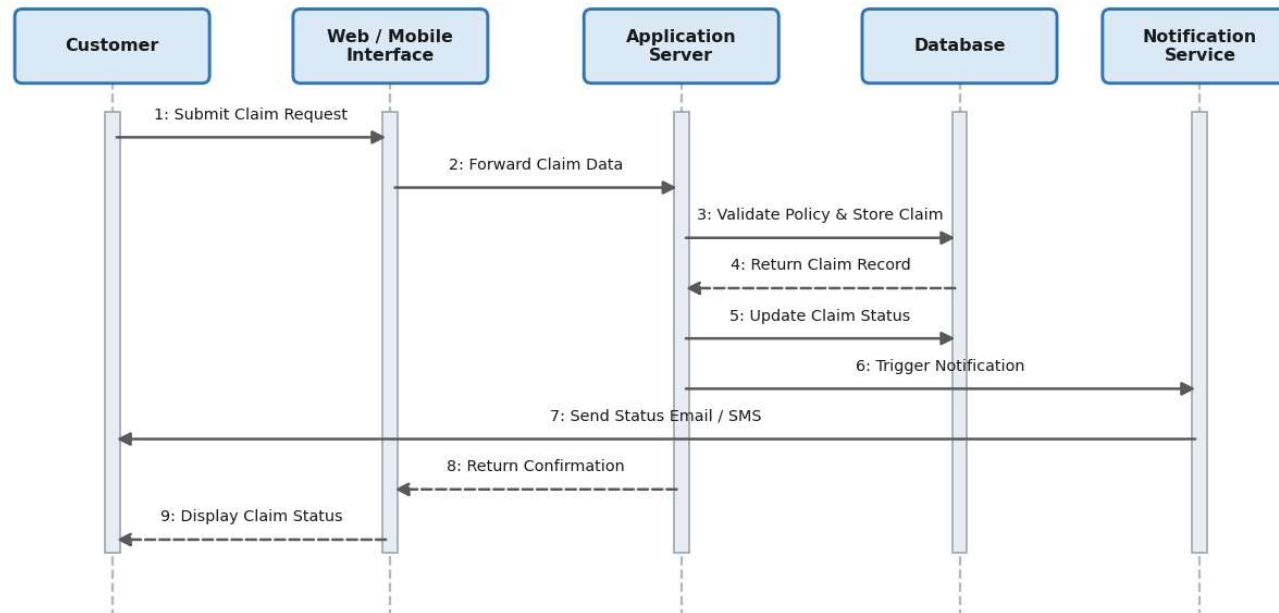


Engineer to Excel

Sequence Diagram



Sequence Diagram — File & Process Insurance Claim





Technology Stack



- **Programming Language:** Java
- **Frameworks:** Spring Boot, Spring MVC
- **Libraries:** Hibernate/JPA, Bootstrap, Chart.js
- **Database:** MySQL
- **Development Tools:** IntelliJ IDEA / Eclipse, Git & GitHub
- **APIs:** Razorpay/Stripe Payment API, Email/SMS Notification API
- **Hardware:** Not Applicable – Web-based Application



Implementation



Completion Status

- **Overall Completion:** 25 %
- **Frontend:** Web interface and user screens developed.
- **Backend:** Java + Spring Boot application structure will be implemented.
- **Database:** MySQL database and entity integration will be implemented.
- **Core Modules:** Policy Recommendation, Claim Management, and Policyholder Dashboard under development/implementation.

Code Summary

- **Java + Spring Boot** used for application development.
- **Spring MVC** handles requests and application flow.
- **Hibernate/JPA** manages database operations.
- **MySQL** stores policyholder, policy, claim, and premium data.



Screenshots



OVERVIEW

Dashboard

My Profile

INSURANCE

Policy Explorer

Smart Recommendation

My Policies

CLAIMS

Submit Claim

Claim Tracking

SERVICES

Documents

Notifications

Help & Support

Voice Assistant

Accessibility

Logout

Search policies, claims, services...

Voice Alerts I Indhu

H

Health Insurance

Healthcare protection category for eligible medical and hospitalization expenses.

₹1,200/month*

Medical Hospital Individual

View Policy

F

Family Health Insurance

Family-oriented health insurance category designed for multiple members and dependents.

₹1,850/month*

Family Medical Dependents

View Policy

L

Life Insurance

Life protection category focused on long-term financial security.

₹1,000/month*

Life Protection Long-term

View Policy

V

Vehicle Insurance

Vehicle protection category for eligible accident and vehicle damage risks.

₹900/month*

Car Accident Damage

View Policy

T

Travel Insurance

Travel protection category for eligible travel-related risks and emergencies.

₹700/month*

Travel Emergency Trip

View Policy

S

Senior Citizen Health

Healthcare-focused category intended for senior citizen requirements.

₹2,200/month*

Senior Health Hospital

View Policy

OVERVIEW

Dashboard

My Profile

INSURANCE

Policy Explorer

Smart Recommendation

My Policies

CLAIMS

Submit Claim

Claim Tracking

SERVICES

Documents

Notifications

Help & Support

Search policies, claims, services...

Voice Alerts

MY INSURANCE

My Policies

View your active insurance coverage.

POLICY	ID	COVERAGE	PREMIUM	STATUS
Health Insurance	HLTH-10245	₹5,00,000	₹1,200/month	ACTIVE
Life Insurance	LIFE-78431	₹5,00,000	₹1,000/month	ACTIVE



Engineer to Excel

UI



Smartinsure

Insurance made smarter.

Smart Insurance Policy Recommendation and Claim Management System for discovering suitable insurance options and managing claims efficiently.

Customer Management

Smart Policy Recommendation

Claim Processing

Voice Accessibility

Welcome Back

Login to your Smartinsure account.

Existing User

New User

Email

Indhu@gmail.com

Password

Login

Invalid email or password.

OVERVIEW

Dashboard

My Profile

INSURANCE

Policy Explorer

Smart Recommendation

My Policies

CLAIMS

Submit Claim

Claim Tracking

SERVICES

Documents

Notifications

Help & Support

Voice Assistant

Accessibility

Logout

Search policies, claims, services...

Voice Alerts I Indhu

Good morning, Indhu

Manage your insurance, discover suitable policies and track your claims.

Find a Policy

Voice Assistant

ACTIVE POLICIES

P

2

Currently active

OPEN CLAIMS

C

0

Requiring attention

TOTAL COVERAGE

₹

₹10L

Across active policies

PROFILE

U

85%

Completion status

Insurance Overview

View all

H

Health Insurance

HLTH-10245 • ₹5L coverage

ACTIVE

L

Life Insurance

LIFE-78431 • ₹5L coverage

ACTIVE

R

Next renewal

Health Insurance • Jan 2027

UPCOMING

Quick Actions

S

Search Policies

Find insurance

R

Get Recommendation

Find best match

C

Submit Claim

Start a claim

D

Documents

View documents

Accessibility



Testing



Unit Tests

- Tested individual functions of Policy Recommendation, Claim Management, and Policyholder Dashboard.
- Verified user registration, policy selection, claim submission, and data validation.
- Checked database operations using MySQL.

Integration Tests

- Tested interaction between Frontend → Java/Spring Boot → MySQL.
- Verified integration of modules and external APIs such as Payment and Email/SMS services.
- Tested end-to-end workflows such as registration → policy purchase → claim filing → dashboard update.

Outputs

- Successful test cases displayed with Pass/Fail status.
- Verified correct data storage, retrieval, and dashboard updates.
- Identified and fixed errors during module and integration testing.



Challenges



Issues

- Managing multiple modules with a centralized database.
- Maintaining secure policyholder and claim information.
- Integrating payment and notification APIs.
- Ensuring accurate real-time dashboard data.

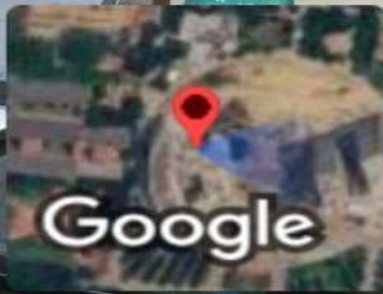
Solutions

- Used **Java + Spring Boot** with a modular architecture.
- Implemented **Hibernate/JPA + MySQL** for structured data management.
- Applied validation and secure authentication mechanisms.
- Tested module integration and data flow systematically.

Next Plan

- Complete remaining module implementation.
- Integrate and test external APIs.
- Perform final system and security testing.
- Complete deployment, documentation, and final presentation.

*Thank
you!*



Eee Department, Mevalurkuppam, Tamil Nadu 602105, India
Lat 13.026599° Long 80.016182°
Wednesday, 19/08/2026 02:46 PM GMT +05:30

